

Mayank Pillai

mayank-pillai.vercel.app | mayank.pillai2024@nst.rishihood.edu.in | linkedin.com/in/mayank-pillai | github.com/mayank-pillai-99 | leetcode.com/nkZEqqcSwd

PROFESSIONAL SUMMARY

Artificial Intelligence undergraduate who builds LLM-powered systems (Text-to-SQL, RAG, LLM evaluation) and deployed full-stack products on Next.js, Node.js and PostgreSQL, with strong algorithmic fundamentals (160+ LeetCode). Seeking AI engineering and full-stack internships.

EDUCATION

Newton School of Technology, Rishihood University

Bachelor of Technology in Artificial Intelligence; CGPA: 9.449/10.0

Sonipat, Haryana

Aug. 2024 – May 2028

- Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Web Development, Computer Networks, Machine Learning

EXPERIENCE

Visiting Summer Research Intern (VISRI)

May 2026 – July 2026

BITS Pilani | Intent-Driven Spatial Search Engine

Pilani, Rajasthan

- Built and deployed a fully offline multilingual spatial search system on a single local GPU, converting conversational voice and text into executable PostGIS SQL through an LLM intent router with a 3-layer FAISS/LangChain RAG fallback, served via a Gradio + Folium map interface
- Designed a Hybrid Evaluation Framework pairing AST-based structural scoring (sqlglot, 7 clause-level dimensions) with live-execution precision/recall, addressing known gaps in exact-match Text-to-SQL benchmarks
- Benchmarked 4 open-weight Text-to-SQL models zero-shot; Qwen2.5-Coder-7B reached a 78.0% pass rate vs. 46.0% (SQLCoder-8B) and 32.0% (CodeS-7B), generalizing to an unseen campus map at 71.2% over 80 queries
- Curated 186 benchmark queries spanning 5 difficulty tiers and 5 Indian languages, using them to select SeamlessM4T v2 (0.470 WER) over Whisper-Large (0.632) for fully offline speech-to-English

PROJECTS

Codebase Copilot | TypeScript, Next.js, Fastify, PostgreSQL (pgvector), Redis, BullMQ, tree-sitter

Sep. 2026

- Built a full-stack tool that ingests any TypeScript/JavaScript GitHub repository and parses it with tree-sitter into a code graph of symbols, imports, calls, and HTTP routes, powering an architecture map, request tracing, and change-impact analysis
- Implemented hybrid retrieval fusing pgvector (HNSW) semantic search with Postgres full-text search via Reciprocal Rank Fusion and one-hop call-graph expansion, streaming answers over SSE with server-validated file/line citations
- Architected a Next.js + Fastify monorepo with a BullMQ/Redis indexing worker and httpOnly-cookie JWT auth, covered by unit, integration, and Playwright tests in GitHub Actions CI and deployed on Vercel and Render

Hitbox | Next.js, React, Express.js, MongoDB, zod, Vitest, GitHub Actions

Dec. 2025

- Built a full-stack social platform for gamers to track, review, and curate lists from IGDB's 500K+ game catalogue, with follows, an activity feed, and comment threads, deployed on Vercel and Render
- Designed explainable recommendation, backlog-ranking, and taste-match engines as pure, testable scoring functions using genre affinity, member ratings, and IGDB time-to-beat data
- Hardened the Express API with zod validation, JWT/bcrypt auth, rate limiting, and helmet headers, backed by 200+ Vitest/Supertest tests in CI and an axe audit with zero WCAG 2.1 AA violations

BookConnect | MongoDB, Express.js, React.js, Node.js, Redux Toolkit, Socket.io, Tailwind CSS

Oct. 2025

- Engineered real-time social networking platform with instant messaging using Socket.io WebSocket connections and event-driven architecture
- Managed global state with Redux Toolkit and built a user discovery feed backed by MongoDB compound indexes and query optimization

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, SQL, HTML, CSS

AI/ML: PyTorch, Hugging Face Transformers, LangChain, FAISS, Gradio; LLM Evaluation, Text-to-SQL, RAG, ASR

Frontend: React.js, Next.js, Tailwind CSS, Redux Toolkit, Context API, React Flow

Backend: Node.js, Express.js, Fastify, Socket.io, BullMQ, Prisma ORM, zod, RESTful APIs, JWT Authentication

Databases: PostgreSQL (pgvector, PostGIS), MongoDB, Redis, MySQL

Tools & Platforms: Git, GitHub Actions, Docker, Vitest, Playwright, Vercel, Render, Postman

Problem Solving: 160+ LeetCode problems solved (Arrays, Trees, Dynamic Programming, Graph Theory)